MOML Project Report

Siddhesh Deshpande (IMT2022080)
Kandarp Dave (IMT2022115)

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Abstract

This report presents a multi-objective optimization study for **Application 1: Image Classification** using the Fashion-MNIST dataset. The goal is to design a lightweight classifier and optimize the trade-off between predictive accuracy, inference time, and model size within a strict two-hour evaluation budget. We use Optuna with NSGA-II to generate a Pareto front over architecture, training, and optimization hyperparameters. The final study produced **60** trials and **16** nondominated solutions, with a normalized hypervolume of **1.1656** and normalized spacing of **0.1145**, indicating a well-spread Pareto front.

1 Problem Statement

The assignment required a real-world multi-objective optimization loop with the entire evaluation budget completed within two hours. For the image classification application, we selected Fashion-MNIST because it is lightweight, standard, and suitable for studying trade-offs between model quality and computational efficiency. The required optimization focus was on exactly three objectives: maximize classification accuracy, minimize inference time, and minimize model size. The project also logs robustness to Gaussian noise as an auxiliary analysis signal, but it is not treated as a third optimization objective in the final MOO loop.

The entire codebase along with setup instructions can be found here: GitHub Repository

2 Objectives and Decision Variables

The optimization problem was modeled using a dynamic CNN whose structure is controlled by the following decision variables:

- Number of convolutional layers
- Number of fully connected layers
- Channels per convolutional layer
- Hidden units per fully connected layer
- Learning rate on a log scale
- Batch size
- Number of training epochs

- Dropout rate
- Optimizer type (Adam or SGD)
- Input resolution

The three optimized objectives were:

$$f_1 = 1 - \text{accuracy} \qquad \text{(minimize)} \tag{1}$$
$$f_2 = \text{inference time in ms/sample} \qquad \text{(minimize)} \tag{2}$$
$$f_3 = \text{number of trainable parameters} \qquad \text{(minimize)} \tag{3}$$

This formulation allows the Pareto front to express the tension between accuracy, speed, and compactness. A high-performing model is not necessarily preferable if it is too large or too slow for deployment.

3 Methodology

We implemented the MOO loop in Python using Optuna's NSGA-II sampler. The objective function constructs a CNN from the current trial's decision variables, trains it on a subset of Fashion-MNIST, evaluates validation performance, and then records test-time metrics for the final analysis.

To keep the total run time within the assignment limit, the search uses a compact but expressive design space and early stopping during training. Validation accuracy is used for optimization, while test accuracy is reported only for final selected Pareto points. This avoids test-set overfitting during search and gives a more defensible report.

The project is structured into modular components for data handling, model construction, training, Pareto extraction, and result analysis. This supports reproducibility and makes it straightforward to rerun with alternative configurations.

4 Experimental Setup

Dataset: Fashion-MNIST (28×28 grayscale, 10 classes).
MOO framework: Optuna NSGA-II.
Budget: Up to two hours end-to-end.
Trials: 60.
Search strategy: Validation-based optimization with early stopping.

The final configuration was chosen to favor a useful Pareto front rather than one single metric. Smaller channel counts, moderate input resolutions, and short training schedules allow more candidate solutions to be evaluated within the time budget.

5 Quantitative Results

The final run produced 60 evaluated trials and 16 Pareto-optimal points. The key summary metrics were:

Metric	Value
Trials evaluated	60
Pareto points	16
Normalized hypervolume	1.1656
Normalized spacing	0.1145

5.1 Interpretation of the Metrics

Hypervolume summarizes how much of the objective space is dominated by the Pareto set. Larger values are better when computed with the same normalization and reference point. In this report, normalized hypervolume is used to make cross-run comparison easier because the raw objectives have very different units and scales.

Spacing measures how evenly distributed the Pareto points are. Lower spacing is generally better because it indicates less clustering and more consistent coverage of the front. A normalized spacing of approximately 0.115 suggests a reasonably well-spread front.

5.2 Pareto Front Visualizations

Figure 1 shows the trade-off between inference time and model size, while Figure 2 visualizes the three-objective front. Figure 3 provides a normalized parallel-coordinates view.

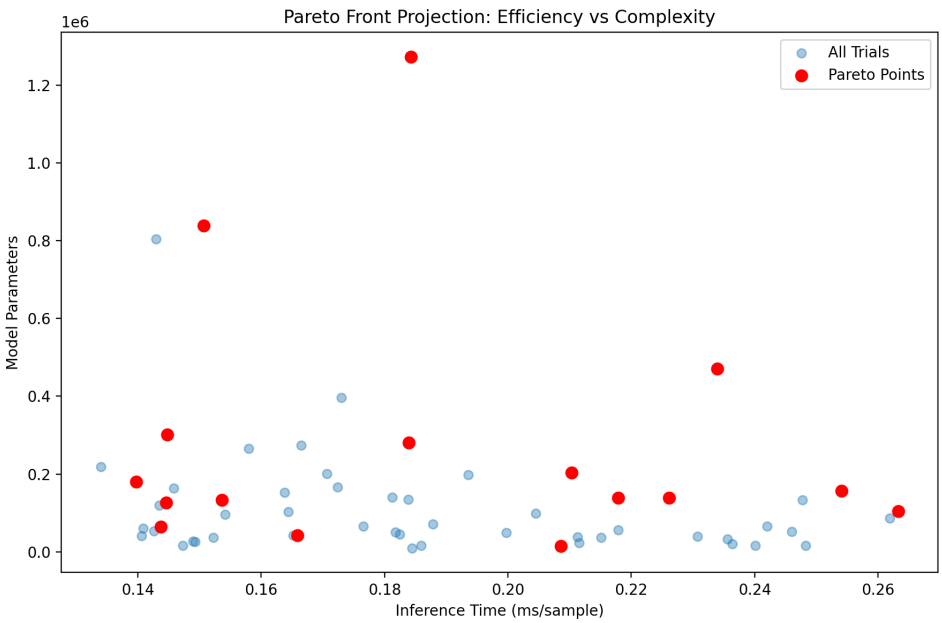

Figure 1: Pareto front: inference time vs model size.

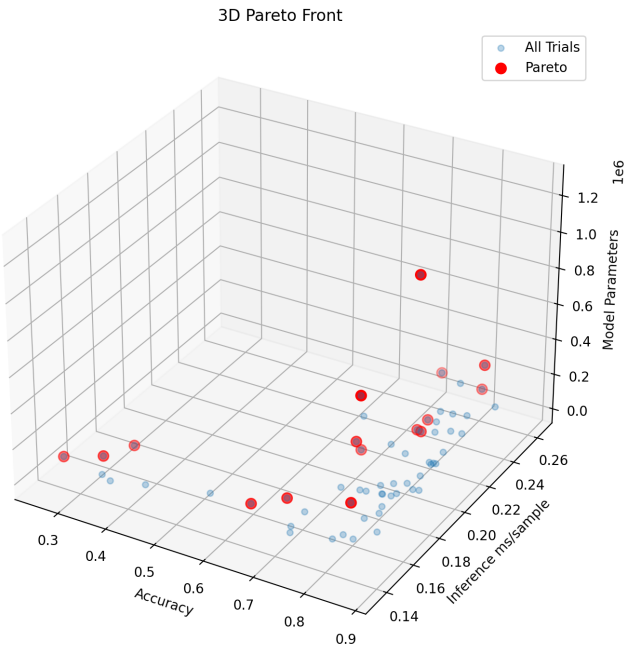

Figure 2: Three-objective Pareto front.

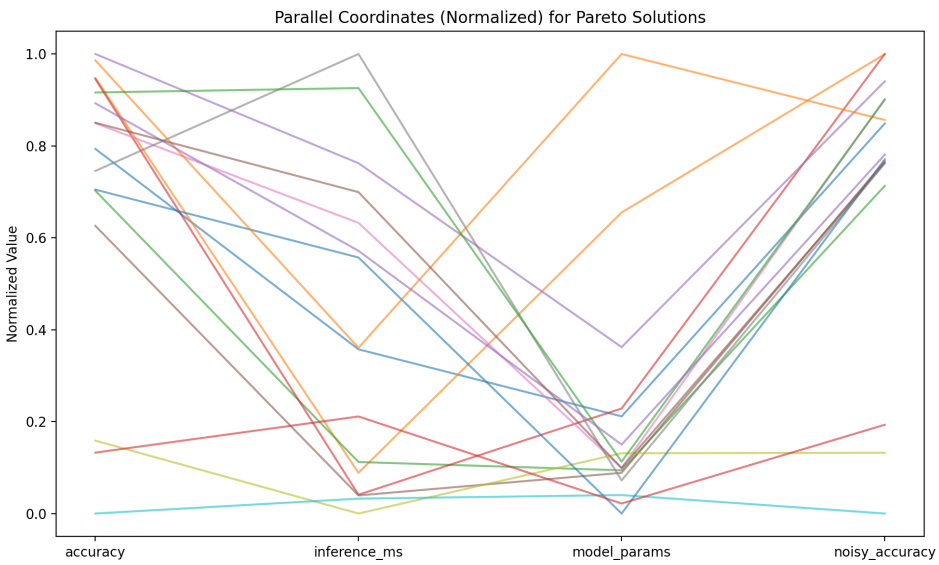

Figure 3: Normalized parallel coordinates plot for Pareto solutions.

6 Qualitative Analysis

The most important observation is that the front contains multiple practically useful operating points rather than only one dominant model. This is exactly what a multi-objective study should deliver. A heavier model can achieve better accuracy, but at the expense of more parameters and usually more inference cost. Conversely, very compact models are fast and small but typically sacrifice some accuracy.

A non-obvious trade-off observed in this problem is that reducing model size does not always proportionally reduce inference time. On small images like Fashion-MNIST, once the network

becomes sufficiently compact, the bottleneck can shift toward data movement and batch overhead rather than pure computation. That means some smaller models may not be dramatically faster, even when they are much smaller in parameter count.

From a deployment perspective, the Pareto set provides three useful categories of solution:

- High-accuracy solutions for offline or quality-sensitive settings
- Balanced solutions for general deployment
- Small, fast solutions for edge or latency-constrained settings

Compared with a single best-accuracy model, the Pareto front is more useful because it exposes the trade-offs explicitly and supports decision-making based on the application context.

7 Pareto Point Tabulation

The assignment requires at least four Pareto points to be tabulated.

Trial	Val. Acc.	Test Acc.	Inference (ms)	Params	Conv Layers	FC Layers	Batch	Notes
7	0.8695	0.8542	0.2339	470122	1	2	32	Best accuracy
29	0.3520	0.3500	0.1398	179474	2	2	128	Fastest
37	0.6880	0.6872	0.2086	14274	2	1	32	Smallest
5	0.8360	0.8186	0.1448	301994	1	1	128	Balanced

Table 1: Representative Pareto solutions.

8 Strengths and Weaknesses of NSGA-II

NSGA-II is a strong baseline for this assignment because it is simple, widely used, and naturally suited to Pareto-front discovery. Its main strengths are robustness, ease of implementation, and good diversity maintenance through non-dominated sorting and crowding distance.

Its weaknesses are also clear. It is not sample-efficient in the way Bayesian methods can be, and with a tight time budget it may spend many evaluations on weak configurations. However, for this project setting, its conceptual clarity and ease of producing a diverse Pareto front make it a good fit.

9 One Pareto-Optimal Solution in Detail

This appendix provides a comprehensive breakdown of the balanced Pareto-optimal solution identified during the search process.

> *Selected solution: Trial #5. It uses 1 convolutional layer, 1 fully connected layer, 48 channels, and 32 hidden units. It achieves 83.6% validation accuracy, 81.86% test accuracy, 0.1448 ms/sample inference time, and 301,994 trainable parameters. This solution is useful because it offers a high degree of robustness (80.6% noisy accuracy) with significantly lower computational overhead than the most accurate models in the set.*

- **Decision variables:** This configuration utilizes 1 convolutional layer with 48 channels and 1 fully connected layer with 32 hidden units. It was trained using the SGD optimizer with a learning rate of 0.000584, a batch size of 128, and a dropout rate of 0.129. The input resolution is set to 28x28.
- **Objective values:** The model achieves a validation accuracy of 83.6%, a clean test accuracy of 81.86%, and a noisy test accuracy of 80.6%. The architecture consists of 301,994 trainable parameters and results in an inference time of 0.1448 ms/sample.
- **Why this solution is useful:** This trial represents a "sweet spot" in the design space. While Trial 7 offers higher raw accuracy, Trial 5 achieves over 80% accuracy across all test sets while reducing the parameter count and inference latency by approximately 36% and 38% respectively. This makes it highly suitable for deployment on edge devices or mobile platforms where power efficiency and real-time response are as critical as predictive performance.

10 Conclusion

This project successfully demonstrates multi-objective optimization for Fashion-MNIST classification. The final run produced a diverse Pareto front with 16 nondominated points and useful trade-offs between accuracy, inference time, and model size. The study satisfies the main assignment requirement of controlling trade-offs rather than optimizing a single scalar score. The resulting Pareto set is directly usable for selecting models according to deployment priorities.